

ORCAS ISLAND, WA, USA

WMO: 722208

Lat: 48.708N

Lon: 122.911W

Elev: 31

StdP: 14.68

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 04224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	26.7	28.4	8.7	8.6	29.5	13.8	11.1	33.6	19.8	47.4	18.1	45.9	6.2	50	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	73.5	62.5	72.3	62.0	70.1	61.3	64.3	71.6	63.1	69.9	62.0	68.4	5.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	81.2	66.3	59.4	75.8	65.4	58.9	74.2	65.0	29.4	72.0	28.6	70.1	27.7	68.6	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	15.2	12.8	21.1	79.9	3.2	2.6	18.7	81.7	16.9	83.2	15.1	84.7	12.7	86.6
			18.4	66.9	3.7	1.7	15.7	68.2	13.5	69.2	11.4	70.2	8.7	71.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	41.8	41.3	44.7	48.5	54.3	58.6	62.6	62.5	57.7	50.9	45.4	40.9	(6)
	DBStd	9.04	5.68	5.54	4.43	3.93	4.14	3.57	3.20	2.85	3.53	4.05	5.70	5.83	(7)
	HDD50	1237	258	247	171	72	11	0	0	0	0	38	155	285	(8)
	HDD65	5199	719	665	630	496	331	194	87	87	220	437	587	746	(9)
	CDD50	1542	5	2	6	26	145	259	392	389	231	65	18	4	(10)
	CDD65	29	0	0	0	0	0	3	14	11	1	0	0	0	(11)
	CDH74	83	0	0	0	0	1	7	45	28	2	0	0	0	(12)
	CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0	(13)
	WSAvg	5.3	5.9	5.5	5.4	5.2	5.0	5.5	5.6	5.1	4.1	4.8	6.1	5.7	(14)
	PrecAvg	25.9	3.7	2.1	2.2	1.7	1.6	1.3	0.6	1.1	1.3	3.0	4.3	3.4	(15)
(16) Precipitation	PrecMax	31.8	5.3	3.9	4.0	3.0	3.0	2.5	1.4	3.6	4.0	7.6	8.2	5.5	(16)
	PrecMin	17.9	1.1	0.1	1.0	0.1	0.5	0.2	0.1	0.1	0.0	0.8	1.2	1.6	(17)
	PrecStd	3.6	1.0	0.9	0.7	0.7	0.8	0.6	0.4	0.9	0.9	1.5	1.8	1.2	(18)
	PrecStd	3.6	1.0	0.9	0.7	0.7	0.8	0.6	0.4	0.9	0.9	1.5	1.8	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	56.9	55.2	57.5	63.4	71.7	74.8	78.8	77.1	72.7	64.1	62.6	57.1	53.8	(19)
	0.4% MCWB	50.7	50.6	50.1	53.9	59.3	62.4	65.0	63.2	62.9	57.8	55.5	51.5	51.5	(20)
	2% DB	54.3	53.7	55.1	59.1	66.2	71.7	74.6	73.3	68.3	61.1	57.1	54.2	54.2	(21)
	2% MCWB	50.2	49.4	49.0	51.5	57.4	61.0	63.0	62.8	60.8	55.4	53.2	50.5	50.5	(22)
	5% DB	52.0	50.2	54.0	56.7	63.8	67.9	72.4	72.1	66.0	58.8	54.9	51.5	51.5	(23)
	5% MCWB	49.3	46.9	48.5	49.9	55.8	59.4	62.0	62.4	59.9	54.2	51.7	47.7	47.7	(24)
	10% DB	49.8	48.2	51.7	54.7	62.6	65.6	70.2	69.9	63.9	57.0	53.8	48.3	48.3	(25)
	10% MCWB	47.2	45.7	47.3	49.0	55.2	58.6	61.2	61.8	58.2	53.0	51.0	46.1	46.1	(26)
	0.4% WB	53.5	52.1	53.3	55.5	60.3	64.1	67.1	65.8	64.6	59.5	57.1	53.8	53.8	(27)
	0.4% MCDB	54.8	54.4	56.4	61.9	67.9	72.4	75.4	73.4	69.9	62.8	58.3	55.7	55.7	(28)
(29) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	51.3	49.5	50.6	53.0	58.2	61.8	64.4	64.3	62.0	57.2	54.4	50.6	50.6	(29)
	2% MCDB	53.3	51.7	53.5	57.4	64.7	68.3	72.3	71.3	66.6	60.0	56.6	53.1	53.1	(30)
	5% WB	49.0	47.6	49.1	51.4	56.8	60.2	62.8	63.3	60.4	55.5	52.6	47.9	47.9	(31)
	5% MCDB	50.8	49.0	51.9	55.3	62.9	66.3	70.4	69.8	64.7	58.1	54.8	49.8	49.8	(32)
	10% WB	46.7	46.2	47.6	50.2	55.4	58.6	61.7	62.2	59.2	54.1	50.6	46.3	46.3	(33)
	10% MCDB	48.3	47.5	50.2	53.9	61.1	64.5	68.8	68.4	63.2	56.5	52.7	47.6	47.6	(34)
(35) Mean Daily Temperature Range	MDBR	7.3	8.7	10.2	11.2	13.0	13.0	14.9	14.5	11.9	9.7	8.3	7.2	7.2	(35)
	5% DB	9.7	10.4	12.2	13.5	15.6	16.6	17.8	17.4	14.9	11.1	9.8	9.2	9.2	(36)
	5% WB	6.0	6.9	7.3	7.7	8.8	8.6	8.9	9.0	8.4	7.1	6.5	6.3	6.3	(37)
	MCDBR	8.8	9.3	11.1	12.0	15.2	16.9	16.1	13.4	10.2	8.8	8.8	8.6	8.6	(38)
	MCWBR	6.4	6.6	7.3	7.7	8.8	8.5	9.0	9.0	8.1	7.3	6.8	6.6	6.6	(39)
	MCWBR	6.4	6.6	7.3	7.7	8.8	8.5	9.0	9.0	8.1	7.3	6.8	6.6	6.6	(39)
(40) Clear-Sky Solar Irradiance	taub	0.319	0.315	0.330	0.357	0.379	0.356	0.351	0.357	0.342	0.332	0.321	0.312	0.312	(40)
	taud	2.520	2.556	2.526	2.443	2.393	2.477	2.519	2.503	2.545	2.559	2.529	2.499	2.499	(41)
	Ebn at Noon	239	270	282	284	280	287	287	280	275	259	236	225	225	(42)
	Edh at Noon	19	23	28	33	36	33	32	31	27	23	19	17	17	(43)
(44) All-Sky Solar Radiation	RadAvg	287	566	875	1337	1721	1872	2056	1737	1227	674	343	230	230	(44)
	RadStd	41	78	107	130	139	189	154	119	102	80	56	31	31	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page